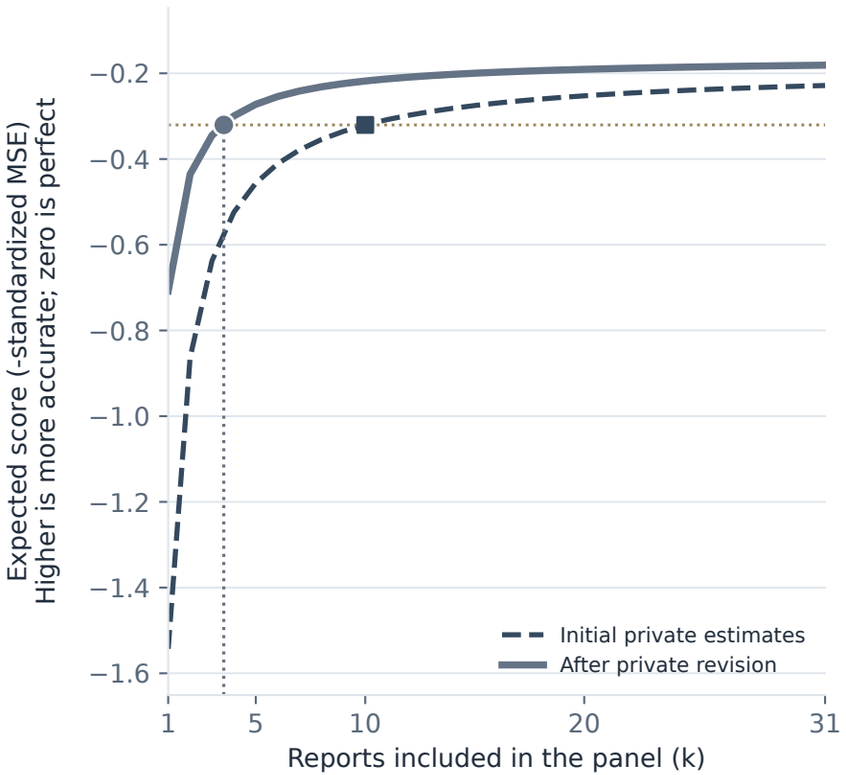
Candy jars: private estimates and social feedback

Six actual jars, with 259-291 candies each. Each curve uses recorded answers from the same people before and after revision.
How many final reports match the accuracy of 10 initial private reports from that same condition?

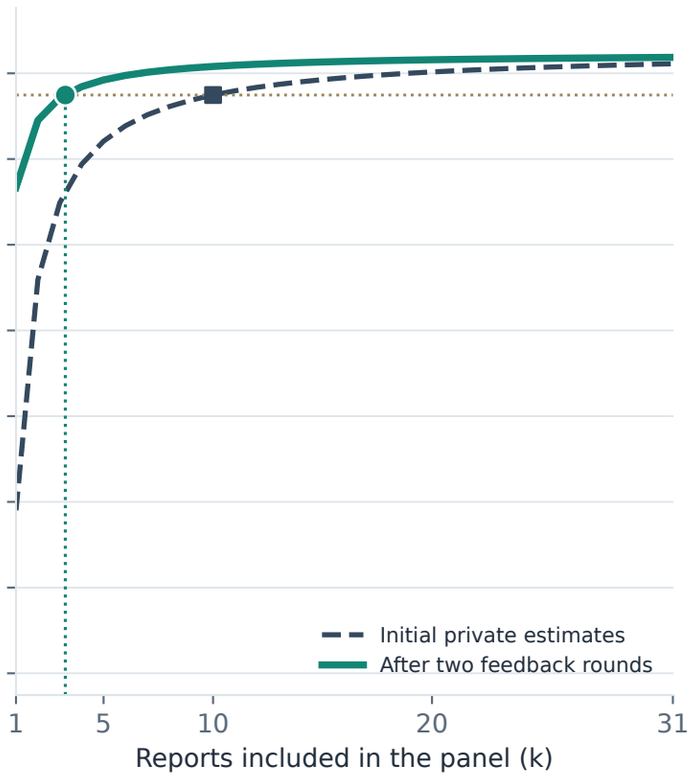
Private throughout



3.54 final reports match 10 initial

Whole reports: 4 | 95% interval: 2 to 8
6 groups; 232 paired person-task records

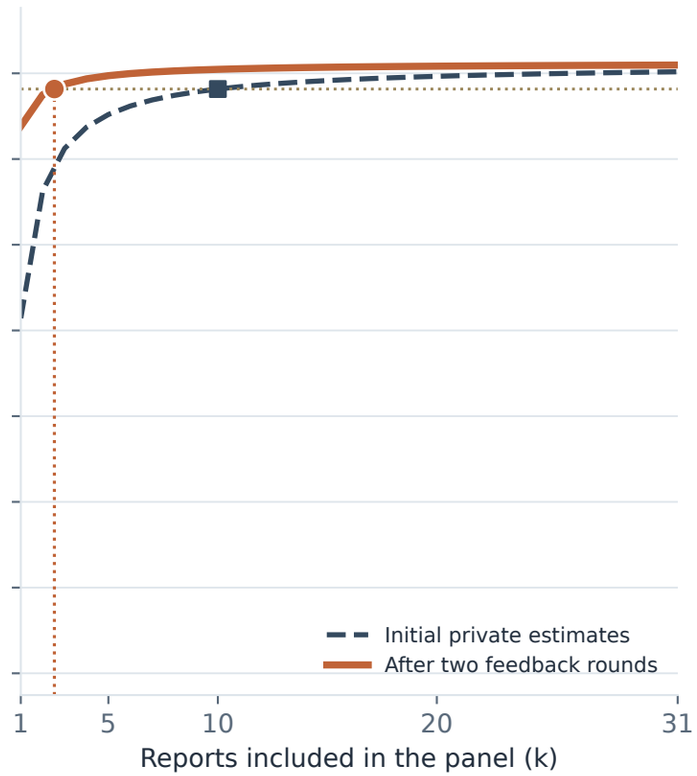
Feedback from several peers



3.25 final reports match 10 initial

Whole reports: 4 | 95% interval: 3 to 10
6 groups; 220 paired person-task records

Feedback dominated by one person



2.53 final reports match 10 initial

Whole reports: 3 | 95% interval: 2 to >31
6 groups; 213 paired person-task records

The dotted horizontal reference is specific to each condition. Fractional crossings interpolate expected score.
Intervals resample whole groups, preserving both rounds and every task; >31 retains draws without a supported crossing.

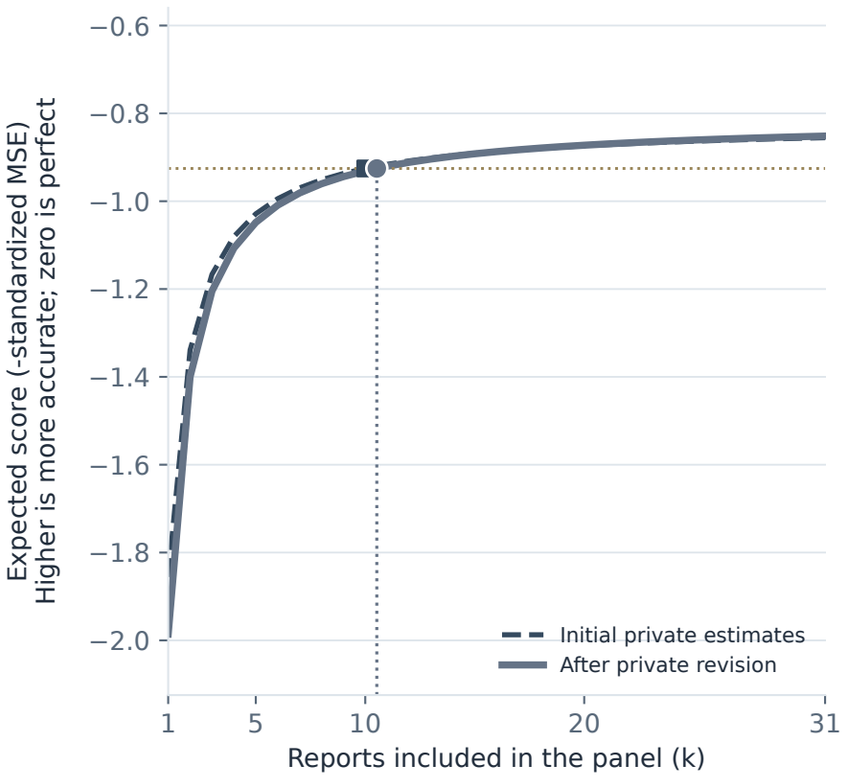
k counts sampled reports from groups originally assigned 40 people. It does not measure the number of people needed to run the discussion.

Source: Becker, Brackbill & Centola (2017) | doi.org/10.1073/pnas.1615978114 | Reanalysis: 28 Sep 2026

The same comparison across all estimation tasks

34 distinct questions in 34 groups. Questions are averaged within groups, then groups receive equal weight within each condition.
How many final reports match the accuracy of 10 initial private reports from that same condition?

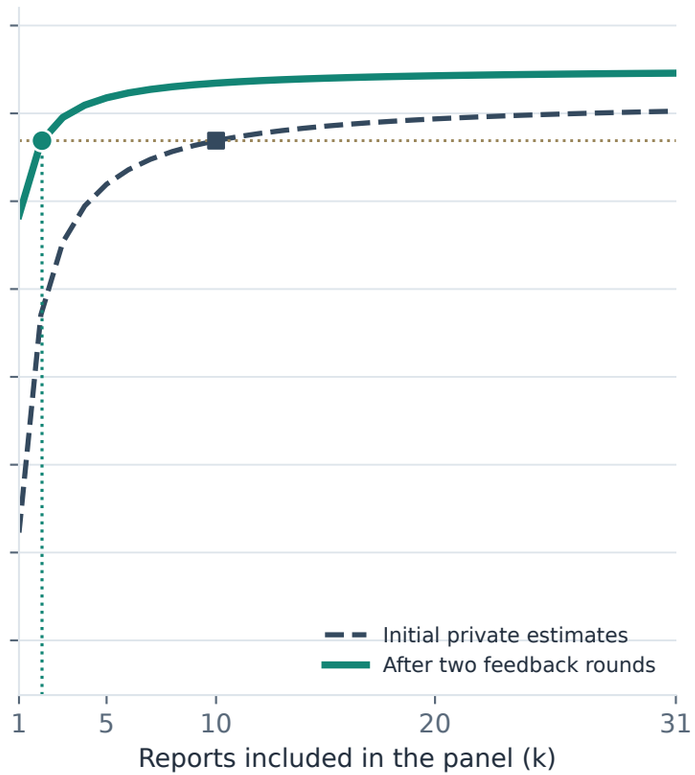
Private throughout



10.53 final reports match 10 initial

Whole reports: 11 | 95% interval: 6 to 22
8 groups; 1,286 paired person-task records

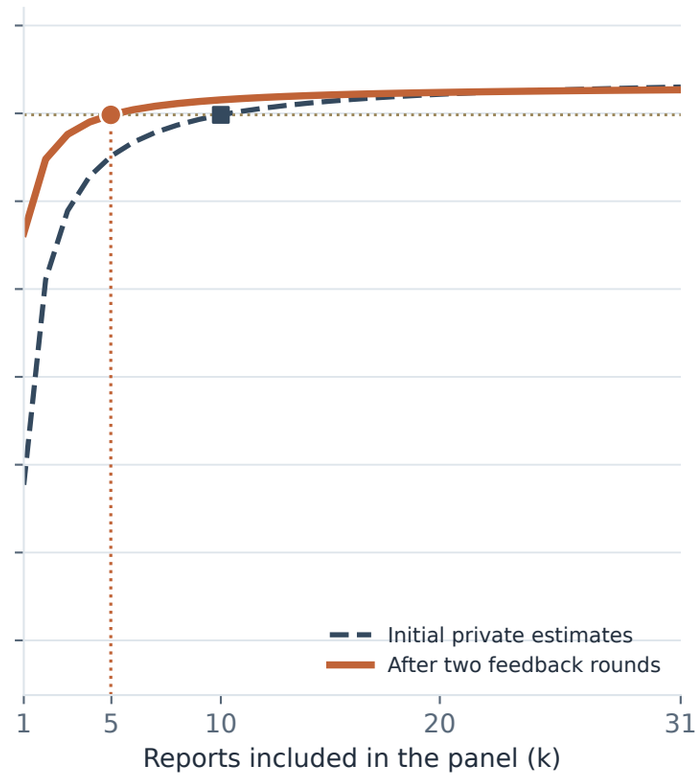
Feedback from several peers



2.05 final reports match 10 initial

Whole reports: 3 | 95% interval: 2 to 3
13 groups; 2,190 paired person-task records

Feedback dominated by one person



4.97 final reports match 10 initial

Whole reports: 5 | 95% interval: 2 to >31
13 groups; 2,150 paired person-task records

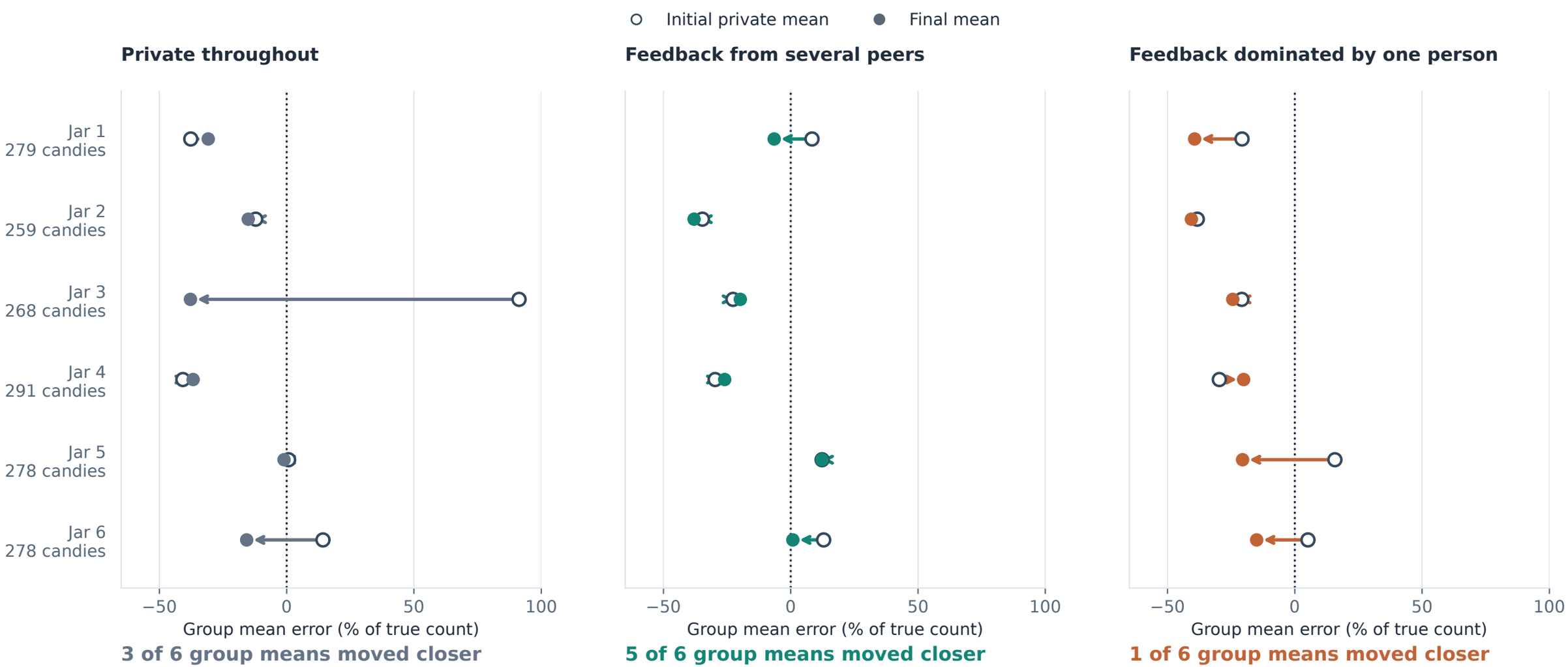
The dotted horizontal reference is specific to each condition. Fractional crossings interpolate expected score.
Intervals resample whole groups, preserving both rounds and every task; >31 retains draws without a supported crossing.

k counts sampled reports from groups originally assigned 40 people. It does not measure the number of people needed to run the discussion.

Source: Becker, Brackbill & Centola (2017) | doi.org/10.1073/pnas.1615978114 | Reanalysis: 28 Sep 2026

Where did each candy-jar estimate move?

Each arrow follows the average estimate of one observed group, using exactly the same respondents before and after revision.
The vertical line marks the true count. Movement toward zero improves the full-group estimate; tighter agreement alone is insufficient.



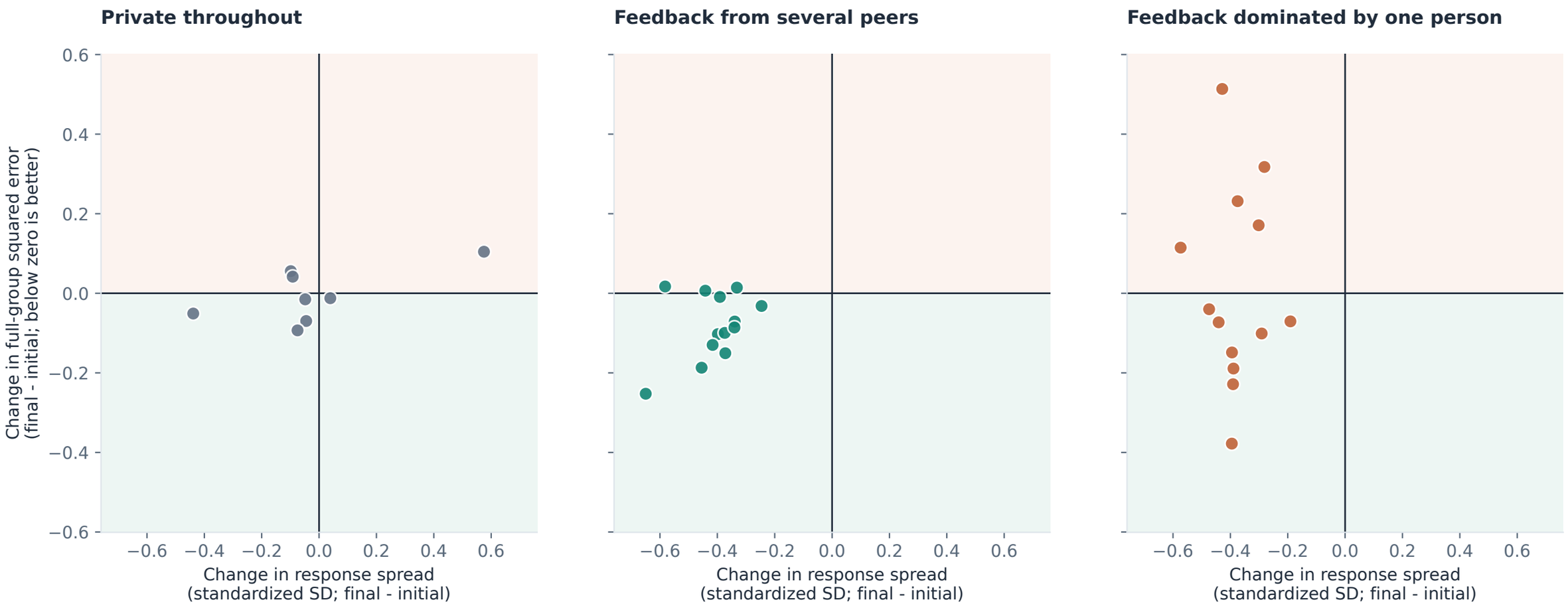
Each row is a different jar; these are 18 separate group-by-jar observations, not a single group changing conditions.
No outlier trimming. One private-control initial mean is elevated by extreme responses; the revision control is therefore essential.

The study used on-screen numerical feedback. It did not test shouting, and the candy brand is not identified here.

Source: Becker, Brackbill & Centola (2017) | doi.org/10.1073/pnas.1615978114 | Reanalysis: 28 Sep 2026

Agreement and accuracy are different outcomes

One point per observed group, averaging its tasks. Both axes compare the same respondents before and after revision.
Left means estimates became more similar. Down means the full-group mean became more accurate.



6 of 8 groups converged

2 converged while error increased

13 of 13 groups converged

3 converged while error increased

13 of 13 groups converged

5 converged while error increased

All 34 observed groups are shown. Group membership, task mix and the released initial-response scale stay fixed between rounds.
Descriptive paired changes; the score is squared error, so this does not reproduce the paper's reported absolute-error effect sizes.

Source: Becker, Brackbill & Centola (2017) | doi.org/10.1073/pnas.1615978114 | Reanalysis: 28 Sep 2026